

LOUISVILLE BOWMAN, KY, USA

WMO: 724235

Lat: 38.228N

Lon: 85.664W

Elev: 165

StdP: 99.36

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13810

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -12.3	(c) -9.1	(d) -19.0	(e) 0.7	(f) -10.2	(g) -16.3	(h) 0.9	(i) -8.0	(j) 9.6	(k) 6.3	(l) 8.6	(m) 4.4	(n) 2.8	(o) 310	(p) 0.424

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.8	(c) 34.1	(d) 24.0	(e) 32.9	(f) 23.7	(g) 31.9	(h) 23.2	(i) 25.8	(j) 31.3	(k) 25.2	(l) 30.6	(m) 24.6	(n) 29.7	(o) 3.2	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.2	19.5	28.5	23.7	18.9	28.1	22.9	18.0	27.4	80.8	31.4	77.8	30.6	75.1	29.9	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.3	(b) 7.3	(c) 6.3	(e) -15.4	(f) 36.3	(g) 4.2	(h) 2.0	(i) -18.4	(j) 37.8	(k) -20.8	(l) 38.9	(m) -23.2	(n) 40.1	(o) -26.2	(p) 41.5		
(a) 8.3	(b) 7.3	(c) 6.3	(e) -16.1	(f) 27.1	(g) 3.9	(h) 1.2	(i) -19.0	(j) 27.9	(k) -21.3	(l) 28.6	(m) -23.5	(n) 29.3	(o) -26.4	(p) 30.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	14.4	1.5	3.8	8.4	14.6	19.6	24.1	25.8	25.4	21.6	15.1	8.7	3.9	(6)
(7)		DBStd	9.85	6.71	6.31	5.88	5.05	4.27	3.03	2.59	2.60	3.88	4.83	5.15	5.79	(7)
(8)		HDD10.0	880	274	191	101	16	1	0	0	0	0	10	85	203	(8)
(9)		HDD18.3	2287	521	408	310	132	38	2	0	0	14	124	291	447	(9)
(10)		CDD10.0	2499	12	16	53	153	299	424	488	478	348	168	46	15	(10)
(11)		CDD18.3	864	0	1	3	19	78	176	230	219	112	24	1	1	(11)
(12)		CDH23.3	8319	0	2	28	162	667	1674	2371	2159	1040	206	8	1	(12)
(13)		CDH26.7	3163	0	0	3	24	188	638	993	880	387	49	1	0	(13)
(14)	Wind	WSAvg	2.6	3.3	3.2	3.1	3.1	2.6	2.3	2.0	1.9	1.9	2.2	2.6	2.9	(14)
(15)	Precipitation	PrecAvg	1168	79	83	117	112	123	99	108	90	88	79	91	99	(15)
(16)		PrecMax	1752	184	268	379	355	294	257	255	223	277	253	194	225	(16)
(17)		PrecMin	829	11	19	26	19	35	12	9	16	2	10	18	16	(17)
(18)		PrecStd	205	40	50	66	66	58	54	54	48	60	53	43	46	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.9	22.4	26.7	29.0	32.0	34.1	36.2	35.7	34.8	30.6	24.2	20.6	(19)
MCWB			15.8	16.1	17.7	18.5	22.6	24.1	25.4	24.0	22.2	21.6	17.9	16.3	(20)	
2%		DB	17.3	19.0	23.6	27.2	30.2	32.8	34.0	33.9	32.6	27.8	22.0	17.7	(21)	
		MCWB	14.4	13.8	15.8	17.8	21.3	23.7	24.7	23.8	22.1	19.7	15.8	14.4	(22)	
5%		DB	14.9	16.7	21.0	25.3	28.9	31.7	32.8	32.5	31.0	25.8	19.3	15.2	(23)	
		MCWB	12.2	12.2	14.2	16.6	20.7	23.1	24.2	23.5	21.7	18.5	14.5	12.2	(24)	
10%		DB	12.2	13.8	18.2	23.2	27.4	30.4	31.5	31.2	28.8	23.3	17.3	12.7	(25)	
		MCWB	9.4	9.8	12.8	16.0	20.3	22.5	23.7	23.0	20.9	17.5	13.3	9.8	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.9	17.3	18.7	20.7	24.0	26.0	27.5	26.7	24.9	23.4	19.0	17.5	(27)
MCD B			18.9	21.5	25.1	26.7	29.1	31.5	32.7	32.3	30.9	28.1	22.2	20.0	(28)	
2%		WB	14.9	15.2	17.0	19.5	22.9	25.2	26.1	25.6	23.9	21.6	17.3	15.1	(29)	
		MCD B	16.9	18.1	21.7	24.4	28.2	30.7	31.7	31.0	29.1	25.7	20.4	17.4	(30)	
5%		WB	12.5	12.8	15.5	18.3	22.0	24.5	25.4	24.9	23.0	20.0	15.4	12.8	(31)	
		MCD B	14.5	15.6	19.8	22.7	27.3	29.7	30.9	30.1	27.8	24.3	18.7	14.7	(32)	
10%		WB	9.5	10.1	13.6	17.1	21.1	23.7	24.8	24.2	22.2	18.4	13.6	10.3	(33)	
		MCD B	12.1	13.3	17.5	21.6	25.5	28.7	29.9	28.9	26.9	22.2	16.8	12.5	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.4	9.2	10.3	11.3	10.4	9.9	9.8	10.2	11.2	11.2	9.9	8.0	(35)
MCDBR			10.9	12.5	13.4	13.5	11.7	11.3	11.1	11.6	13.4	13.5	12.6	10.9	(36)	
MCWBR			8.7	9.1	7.9	6.8	5.1	4.4	3.9	4.1	4.9	6.4	8.1	8.6	(37)	
5% WB		MCDBR	9.9	11.0	11.2	10.4	10.0	9.8	9.7	9.7	10.2	10.9	10.7	9.7	(38)	
		MCWBR	8.6	9.4	7.6	6.5	5.1	4.4	4.2	4.0	4.6	6.1	8.2	8.7	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.315	0.329	0.359	0.403	0.442	0.455	0.479	0.476	0.421	0.362	0.330	0.308	(40)	
taud		2.458	2.390	2.381	2.259	2.190	2.188	2.131	2.131	2.260	2.427	2.465	2.482	(41)		
Ebn at Noon		865	896	901	876	844	830	806	799	824	847	838	847	(42)		
Edh at Noon		82	100	111	133	146	146	153	150	124	95	81	75	(43)		
(44)	All-Sky Solar Radiation	RadAvg	1.91	2.62	3.70	4.91	5.47	6.22	5.90	5.52	4.54	3.37	2.31	1.65	(44)	
(45)		RadStd	0.20	0.36	0.35	0.34	0.49	0.35	0.36	0.32	0.43	0.36	0.26	0.16	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+126	+85
(47)	Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+120	+77

Nomenclature: See separate page